

CS422 Assignment 1

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Implementation details

We have done Basic Block level instrumentation for counting number of instructions to speedup the fast forward phase. Terminate check is also placed at basic block level. In a basic block, for each instruction, we check the fast forward condition and insert the analysis call accordingly.

We have made a struct for counting different types of instructions in part A.

For part B, the CPI can be directly calculated using the values of part A.

For part C, we have maintained two bool type vectors, `instr_mem` and `data_mem` to count the different instruction and data addresses touched during the entire program, at the granularity of 32 bytes, since $4\text{ GB} \gg 5 = 134217728$, we set the sizes of those vectors to $134217728 + 2$ (adding 2 for safety to overflow), so that each operation is $O(1)$, and we use only 16 MB of memory for each array, since due to internal C++ optimization each element of a bool type vector takes 1 bit instead of 1 byte.

For part D, we have maintained vectors and some global variables to store the observed counts.

Usage

The following commands are used to build and run the tool on a benchmark application:

Build

```
make TARGET=ia32 obj-ia32/HW1.so
```

Run Tool on a Benchmark Application

```
navigate to benchmark directory: cd /path/to/spec_2006/<benchmark>/  
command: /path/to/pin -t /path/to/obj-ia32/HW1.so -f <fast_forward_count>  
         -o <output_filename> -- ./<benchmark_executable> <command_line_args>
```

Contents

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1 400.perlbench

Part A: Instruction Type Results

Instruction Type	Count	Percentage (%)
Loads	357006003	22.852000
Stores	205414771	13.148600
NOPs	946254	0.060570
Direct calls	12647368	0.809560
Indirect calls	2861038	0.183136
Returns	15508408	0.992696
Unconditional branches	30572441	1.956950
Conditional branches	129939846	8.317470
Logical operations	99892892	6.394160
Rotate and shift	4307390	0.275717
Flag operations	856201	0.054806
Vector instructions	0	0.000000
Conditional moves	0	0.000000
MMX and SSE instructions	0	0.000000
System calls	0	0.000000
Floating point instructions	914580	0.058542
Others	701384418	44.895700

Part B: CPI Results CPI : 25.840500

Part C: Memory Footprint Results

Instruction Footprint (32 byte Blocks) : 2833

Instruction Footprint (Bytes) : 90656

Data Footprint (32 byte Blocks) : 31138

Data Footprint (Bytes) : 996416

Single Instruction Block Proportion : 0.891957

Multiple Instruction Block Proportion : 0.108043

Single Data Block Access Proportion : 0.889813

Multiple Data Block Access Proportion : 0.110187

Part D: Instruction Analysis Results

Instruction Size Results

Instruction Size (Bytes)	Count
1	117138587
2	256450139
3	274763605
4	52825971
5	78533059
6	186289505
7	33999094
8	28
9	0
10	12
11	0
12	0
13	0
14	0
15	0

Instruction Operand Results

Number of Instruction Operands	Count
0	946254
1	1069415
2	521095832
3	355202459
4	102872618
5	15510035
6	3303387

Instruction Register Read Operand Results

Number of Register Read Operands	Count
0	9973765
1	260698739
2	537063327
3	179151963
4	7182864
5	2625955
6	3303387

Instruction Register Write Operand Results

Number of Register Write Operands	Count
0	135594669
1	685366179
2	175732349
3	2402580
4	904223

Memory Instruction Operand Results

Number of Memory Operands	Count
0	453567121
1	540302308
2	5961407

Memory Instruction Read Operand Results

Number of Memory Read Operands	Count
0	643807446
1	355119190
2	904200

Memory Instruction Write Operand Results

Number of Memory Write Operands	Count
0	794873340
1	204957496

Maximum Number of Bytes Touched : 8
Average Number of Bytes Touched : 2.041890
Maximum Value of Immediate : 2147483647
Minimum Value of Immediate : -2147483648
Maximum Value of Displacement : 135918120
Minimum Value of Displacement : -1408

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Part A: Instruction Type Results

Instruction Type	Count	Percentage (%)
Loads	452706087	26.884800
Stores	231174121	13.728700
NOPs	36514	0.002168
Direct Calls	791598	0.047011
Indirect Calls	13	0.000001
Returns	791607	0.047011
Unconditional Branches	21299224	1.264890
Conditional Branches	129923115	7.715720
Logical Operations	71000751	4.216510
Rotate and Shift	61831995	3.672000
Flag Operations	6130	0.000364
Vector Instructions	0	0.000000
Conditional Moves	0	0.000000
MMX and SSE Instructions	0	0.000000
System Calls	0	0.000000
Floating Point Instructions	0	0.000000
Others	714314968	42.420900

Part B: CPI Results CPI : 29.023300

Part C: Memory Footprint Results

Instruction Footprint (32 byte Blocks) : 759

Instruction Footprint (Bytes) : 24288

Data Footprint (32 byte Blocks) : 2449695

Data Footprint (Bytes) : 78390240

Single Instruction Block Proportion : 0.887865

Multiple Instruction Block Proportion : 0.112135

Single Data Block Access Proportion : 0.843361

Multiple Data Block Access Proportion : 0.156639

Part D: Instruction Analysis Results

Instruction Size Results

Instruction Size (Bytes)	Count
1	38611271
2	219201473
3	436965697
4	75326976
5	22047450
6	141357273
7	51341618
8	15085632
9	0
10	62610
11	0
12	0
13	0
14	0
15	0

Instruction Operand Results

Number of Instruction Operands	Count
0	36514
1	6147
2	597647496
3	382870389
4	2693849
5	14191397
6	2554208

Instruction Register Read Operand Results

Number of Register Read Operands	Count
0	5189068
1	183351701
2	533608178
3	215105829
4	46791217
5	13399799
6	2554208

Instruction Register Write Operand Results

Number of Register Write Operands	Count
0	132331436
1	712782402
2	152331937
3	2554225

Memory Instruction Operand Results

Number of Memory Operands	Count
0	399852542
1	597593235
2	2550138

Memory Instruction Read Operand Results

Number of Memory Read Operands	Count
0	547289828
1	452706087

Memory Instruction Write Operand Results

Number of Memory Write Operands	Count
0	768821794
1	231174121

Maximum Number of Bytes Touched : 8
Average Number of Bytes Touched : 2.091750
Maximum Value of Immediate : 1431655766
Minimum Value of Immediate : -858993459
Maximum Value of Displacement : 135000192
Minimum Value of Displacement : -4848

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Part A: Instruction Type Results

Instruction Type	Count	Percentage (%)
Loads	338875493	22.453800 %
Stores	170334887	11.286400 %
NOPs	3906091	0.258817 %
Direct calls	30224558	2.002670 %
Indirect calls	396025	0.026241 %
Returns	30620582	2.028910 %
Unconditional branches	31771972	2.105200 %
Conditional branches	145812580	9.661510 %
Logical operations	54594487	3.617420 %
Rotate and shift	634876	0.042067 %
Flag operations	0	0.000000 %
Vector instructions	0	0.000000 %
Conditional moves	0	0.000000 %
MMX and SSE instructions	0	0.000000 %
System calls	0	0.000000 %
Floating point instructions	0	0.000000 %
Others	702038829	46.517000 %

Part B: CPI Results CPI : 24.280700

Part C: Memory Footprint Results

Instruction Footprint (32 byte Blocks) : 244

Instruction Footprint (Bytes) : 7808

Data Footprint (32 byte Blocks) : 69172

Data Footprint (Bytes) : 2213504

Single Instruction Block Proportion : 0.932853

Multiple Instruction Block Proportion : 0.067147

Single Data Block Accesses Proportion : 0.935233

Multiple Data Block Accesses Proportion : 0.064767

Part D: Instruction Analysis Results

Instruction Size Results

Instruction Size (Bytes)	Count
1	181997507
2	305572205
3	282643616
4	87585846
5	53447844
6	76319268
7	12433714

Instruction Operand Results

Number of Instruction Operands	Count
0	3906091
1	2490686
2	426140831
3	363398130
4	173443679
5	30620583

Instruction Register Read Operand Results

Number of Register Read Operands	Count
0	7270571
1	317012590
2	482051280
3	189700827
4	3568707
5	396025

Instruction Register Write Operand Results

Number of Register Write Operands	Count
0	67550307
1	724663404
2	207786289

Memory Instruction Operand Results

Number of Memory Operands	Count
0	495947400
1	503656575
2	396025

Memory Instruction Read Operand Results

Number of Memory Read Operands	Count
0	661124507
1	338875493

Memory Instruction Write Operand Results

Number of Memory Write Operands	Count
0	829665113
1	170334887

Maximum Number of Bytes Touched : 8

Average Number of Bytes Touched : 1.900170

Maximum Value of Immediate : 136599840

Minimum Value of Immediate : -2147483587

Maximum Value of Displacement : 138633660

Minimum Value of Displacement : -96

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Part A: Instruction Type Results

Instruction Type	Count	Percentage (%)
Loads	415213373	27.222600
Stores	110039541	7.214510
NOPs	1477640	0.096878
Direct calls	12556568	0.823245
Indirect calls	0	0.000000
Returns	12556567	0.823245
Unconditional branches	8314430	0.545118
Conditional branches	178242886	11.686100
Logical operations	75119465	4.925050
Rotate and shift	3516420	0.230547
Flag operations	0	0.000000
Vector instructions	0	0.000000
Conditional moves	0	0.000000
MMX and SSE instructions	0	0.000000
System calls	0	0.000000
Floating point instructions	0	0.000000
Others	708216024	46.432700

Part B: CPI Results CPI : 24.761600

Part C: Memory Footprint Results

Instruction Footprint (32 byte Blocks) : 65

Instruction Footprint (Bytes) : 2080

Data Footprint (32 byte Blocks) : 11673113

Data Footprint (Bytes) : 373539616

Single Instruction Block Proportion : 0.925908

Multiple Instruction Block Proportion : 0.074092

Single Data Block Access Proportion : 0.945577

Multiple Data Block Access Proportion : 0.054423

Part D: Instruction Analysis Results

Instruction Size Results

Instruction Size (Bytes)	Count
1	80626108
2	485394962
3	315526941
4	50531706
5	22076054
6	5249475
7	40594754
8	0
9	0
10	0
11	0
12	0
13	0
14	0
15	0

Instruction Operand Results

Number of Instruction Operands	Count
0	1477640
1	0
2	484783817
3	457347005
4	43834970
5	12556568

Instruction Register Read Operand Results

Number of Register Read Operands	Count
0	3759790
1	148509144
2	677533368
3	170197698

Instruction Register Write Operand Results

Number of Register Write Operands	Count
0	70905928
1	770283430
2	158772130
3	38512

Memory Instruction Operand Results

Number of Memory Operands	Count
0	487181825
1	512818175
2	0

Memory Instruction Read Operand Results

Number of Memory Read Operands	Count
0	584786627
1	415213373

Memory Instruction Write Operand Results

Number of Memory Write Operands	Count
0	889960459
1	110039541

Maximum Number of Bytes Touched : 4

Average Number of Bytes Touched : 2.051270

Maximum Value of Immediate : 1374389535

Minimum Value of Immediate : -100000000

Maximum Value of Displacement : 134957120

Minimum Value of Displacement : -76

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Part A: Instruction Type Results

Instruction Type	Count	Percentage (%)
Loads	546247389	33.148900
Stores	101612623	6.166340
NOPs	11471	0.000696
Direct calls	3199444	0.194158
Indirect calls	135	0.000008
Returns	3199579	0.194166
Unconditional branches	13383523	0.812176
Conditional branches	103025352	6.252070
Logical operations	13717300	0.832431
Rotate and shift	10236564	0.621203
Flag operations	23458139	1.423550
Vector instructions	0	0.000000
Conditional moves	0	0.000000
MMX and SSE instructions	0	0.000000
System calls	0	0.000000
Floating point instructions	306634751	18.608100
Others	523133742	31.746200

Part B: CPI Results CPI : 28.127500

Part C: Memory Footprint Results

Instruction Footprint (32 byte Blocks) : 1180

Instruction Footprint (Bytes) : 37760

Data Footprint (32 byte Blocks) : 5694020

Data Footprint (Bytes) : 182208640

Single Instruction Block Proportion : 0.920013

Multiple Instruction Block Proportion : 0.079987

Single Data Block Access Proportion : 0.849143

Multiple Data Block Access Proportion : 0.150857

Part D: Instruction Analysis Results

Instruction Size Results

Instruction Size (Bytes)	Count
1	76908840
2	442246665
3	398654817
4	16506344
5	3277725
6	40938711
7	17528236
8	3938662
9	0
10	0
11	0
12	0
13	0
14	0
15	0

Instruction Operand Results

Number of Instruction Operands	Count
0	11471
1	23
2	415479217
3	394886789
4	186386703
5	3235797

Instruction Register Read Operand Results

Number of Register Read Operands	Count
0	23668532
1	216024139
2	576812999
3	140942988
4	42515010
5	36332

Instruction Register Write Operand Results

Number of Register Write Operands	Count
0	69709095
1	765054010
2	165236895

Memory Instruction Operand Results

Number of Memory Operands	Count
0	529935585
1	470064287
2	128

Memory Instruction Read Operand Results

Number of Memory Read Operands	Count
0	582771191
1	417228809

Memory Instruction Write Operand Results

Number of Memory Write Operands	Count
0	916237789
1	83762211

Maximum Number of Bytes Touched : 8

Average Number of Bytes Touched : 2.467730

Maximum Value of Immediate : 2147483647

Minimum Value of Immediate : -1074790400

Maximum Value of Displacement : 135856736

Minimum Value of Displacement : -344

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Part A: Instruction Type Results

Instruction Type	Count	Percentage (%)
Loads	547670944	33.736700
Stores	75699025	4.663080
NOPs	34317	0.002114
Direct calls	144578	0.008906
Indirect calls	937	0.000058
Returns	145515	0.008964
Unconditional branches	205820	0.012679
Conditional branches	144361223	8.892690
Logical operations	1158479	0.071363
Rotate and shift	294092	0.018116
Flag operations	5648	0.000348
Vector instructions	0	0.000000
Conditional moves	0	0.000000
MMX and SSE instructions	0	0.000000
System calls	0	0.000000
Floating point instructions	40212	0.002477
Others	853608216	52.582500

Part B: CPI Results CPI : 27.495800

Part C: Memory Footprint Results

Instruction Footprint (32 byte Blocks) : 462

Instruction Footprint (Bytes) : 14784

Data Footprint (32 byte Blocks) : 85346

Data Footprint (Bytes) : 2731072

Single Instruction Block Proportion : 0.915272

Multiple Instruction Block Proportion : 0.084728

Single Data Block Access Proportion : 0.888579

Multiple Data Block Access Proportion : 0.111421

Part D: Instruction Analysis Results

Instruction Size Results

Instruction Size (Bytes)	Count
1	25077914
2	302860246
3	296150925
4	270389432
5	24861194
6	68360442
7	416469
8	11883378
9	0
10	0
11	0
12	0
13	0
14	0
15	0

Instruction Operand Results

Number of Instruction Operands	Count
0	34887
1	3491
2	566127646
3	432938017
4	712697
5	159485
6	23777

Instruction Register Read Operand Results

Number of Register Read Operands	Count
0	610915
1	75619798
2	562398523
3	281426030
4	79906050
5	14907
6	23777

Instruction Register Write Operand Results

Number of Register Write Operands	Count
0	75149763
1	755279808
2	169546652
3	23777

Memory Instruction Operand Results

Number of Memory Operands	Count
0	376807599
1	623167687
2	23751

Memory Instruction Read Operand Results

Number of Memory Read Operands	Count
0	452335976
1	547663061

Memory Instruction Write Operand Results

Number of Memory Write Operands	Count
0	924304648
1	75694389

Maximum Number of Bytes Touched : 8
Average Number of Bytes Touched : 2.491220
Maximum Value of Immediate : 2147483647
Minimum Value of Immediate : -987654321
Maximum Value of Displacement : 135294312
Minimum Value of Displacement : -580

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Part A: Instruction Type Results

Instruction Type	Count	Percentage (%)
Loads	371208366	23.186400
Stores	229766215	14.351600
NOPs	803454	0.050185
Direct calls	21322890	1.331870
Indirect calls	3688048	0.230363
Returns	25010938	1.562230
Unconditional branches	22183560	1.385630
Conditional branches	117345375	7.329620
Logical operations	60018824	3.748890
Rotate and shift	7153668	0.446832
Flag operations	20154254	1.258870
Vector instructions	0	0.000000
Conditional moves	0	0.000000
MMX and SSE instructions	0	0.000000
System calls	0	0.000000
Floating point instructions	96925334	6.054150
Others	625393655	39.063300

Part B: CPI Results

CPI : 26.901300

Part C: Memory Footprint Results

Instruction Footprint (32 byte Blocks) : 901

Instruction Footprint (Bytes) : 28832

Data Footprint (32 byte Blocks) : 553907

Data Footprint (Bytes) : 17725024

Single Instruction Block Proportion : 0.927352

Multiple Instruction Block Proportion : 0.072648

Single Data Block Accesses Proportion : 0.894120

Multiple Data Block Accesses Proportion : 0.105880

Part D: Instruction Analysis Results

Instruction Size Results

Instruction Size (Bytes)	Count
1	154553896
2	308534239
3	382110382
4	34342996
5	45112742
6	48829664
7	26515660
10	421

Instruction Operand Results

Number of Instruction Operands	Count
0	803454
1	225834
2	518299002
3	281796989
4	172711322
5	25024529
6	1138870

Instruction Register Read Operand Results

Number of Register Read Operands	Count
0	29001600
1	197519806
2	540435426
3	219759339
4	8459554
5	3685405
6	1138870

Instruction Register Write Operand Results

Number of Register Write Operands	Count
0	165343555
1	665540338
2	167601343
3	375894
4	1138870

Memory Instruction Operand Results

Number of Memory Operands	Count
0	449279644
1	545896081
2	4824275

Memory Instruction Read Operand Results

Number of Memory Read Operands	Count
0	661713281
1	337147849
2	1138870

Memory Instruction Write Operand Results

Number of Memory Write Operands	Count
0	780939796
1	219060204

Maximum Number of Bytes Touched : 8
Average Number of Bytes Touched : 2.324920
Maximum Value of Immediate : 2147483647
Minimum Value of Immediate : -2092037281
Maximum Value of Displacement : 136117612
Minimum Value of Displacement : -104

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Part A: Instruction Type Results

Instruction Type	Count	Percentage (%)
Loads	357436990	23.627500
Stores	155774956	10.297100
NOPs	26112489	1.726100
Direct Calls	11530299	0.762183
Indirect Calls	8838388	0.584240
Returns	20372746	1.346690
Unconditional Branches	7564639	0.500042
Conditional Branches	191230020	12.640800
Logical Operations	34154916	2.257730
Rotate and Shift	4926457	0.325652
Flag Operations	1448098	0.095723
Vector Instructions	0	0.000000
Conditional Moves	0	0.000000
MMX and SSE Instructions	0	0.000000
System Calls	0	0.000000
Floating Point Instructions	6322761	0.417951
Others	687087513	45.418300

Part B: CPI Results CPI : 24.408000

Part C: Memory Footprint Results

Instruction Footprint (32 byte Blocks) : 2286

Instruction Footprint (Bytes) : 73152

Data Footprint (32 byte Blocks) : 890839

Data Footprint (Bytes) : 28506848

Single Instruction Block Proportion : 0.933089

Multiple Instruction Block Proportion : 0.066911

Single Data Block Access Proportion : 0.886010

Multiple Data Block Access Proportion : 0.113990

Part D: Instruction Analysis Results

Instruction Size Results

Instruction Size (Bytes)	Count
1	143612757
2	332671065
3	445804190
4	28414136
5	21378395
6	21063253
7	6718936
8	229735
9	51390
10	56143
11	0
12	0
13	0
14	0
15	0

Instruction Operand Results

Number of Instruction Operands	Count
0	26112489
1	648405
2	409723397
3	432925374
4	91910243
5	22067590
6	16612502

Instruction Register Read Operand Results

Number of Register Read Operands	Count
0	32925875
1	245145155
2	446314783
3	249204346
4	1061554
5	8735785
6	16612502

Instruction Register Write Operand Results

Number of Register Write Operands	Count
0	109059576
1	695140656
2	179185311
3	16614457

Memory Instruction Operand Results

Number of Memory Operands	Count
0	517109651
1	457556123
2	24922552

Memory Instruction Read Operand Results

Number of Memory Read Operands	Count
0	644175082
1	355413244

Memory Instruction Write Operand Results

Number of Memory Write Operands	Count
0	844800696
1	154787630

Maximum Number of Bytes Touched : 8

Average Number of Bytes Touched : 2.017160

Maximum Value of Immediate : 2147483647

Minimum Value of Immediate : -1431655765

Maximum Value of Displacement : 139657912

Minimum Value of Displacement : -1392